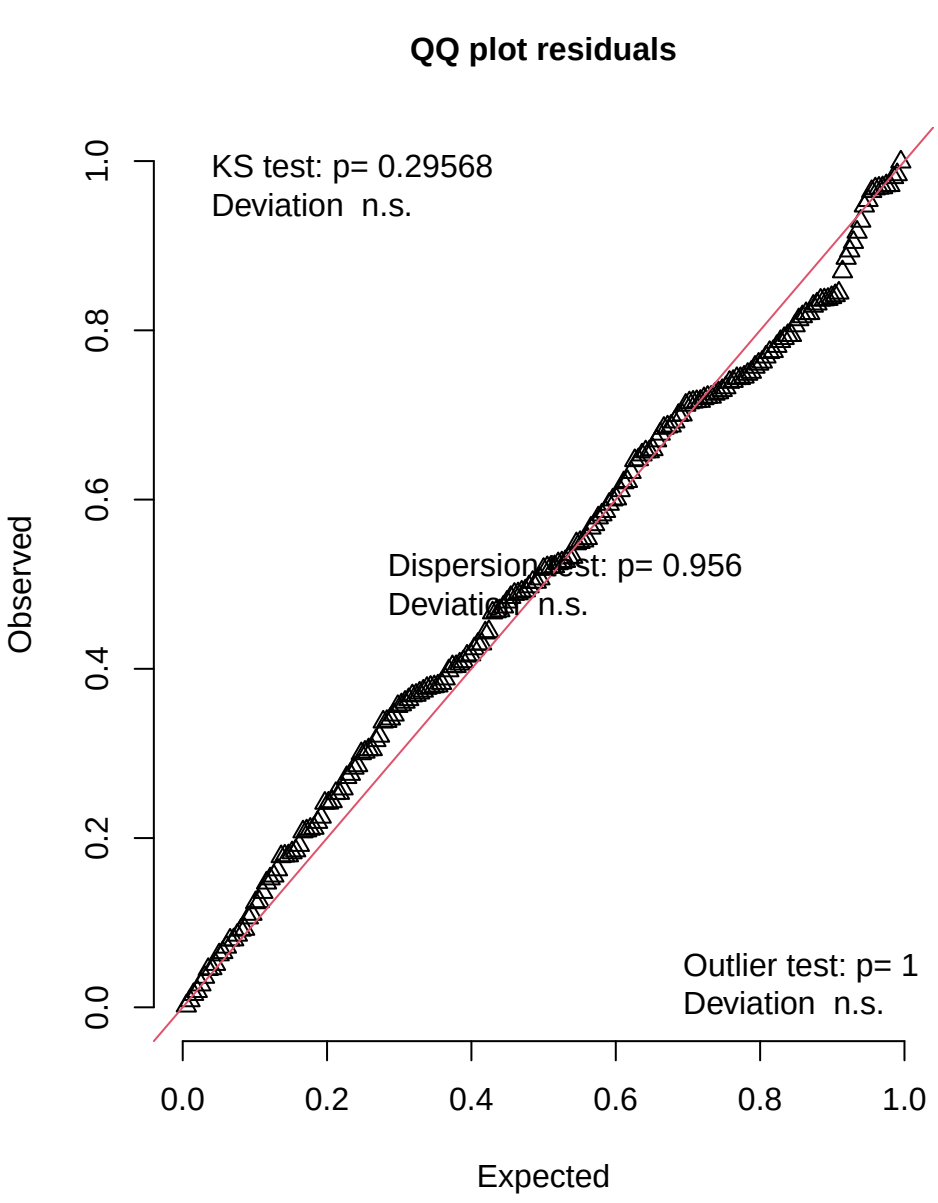
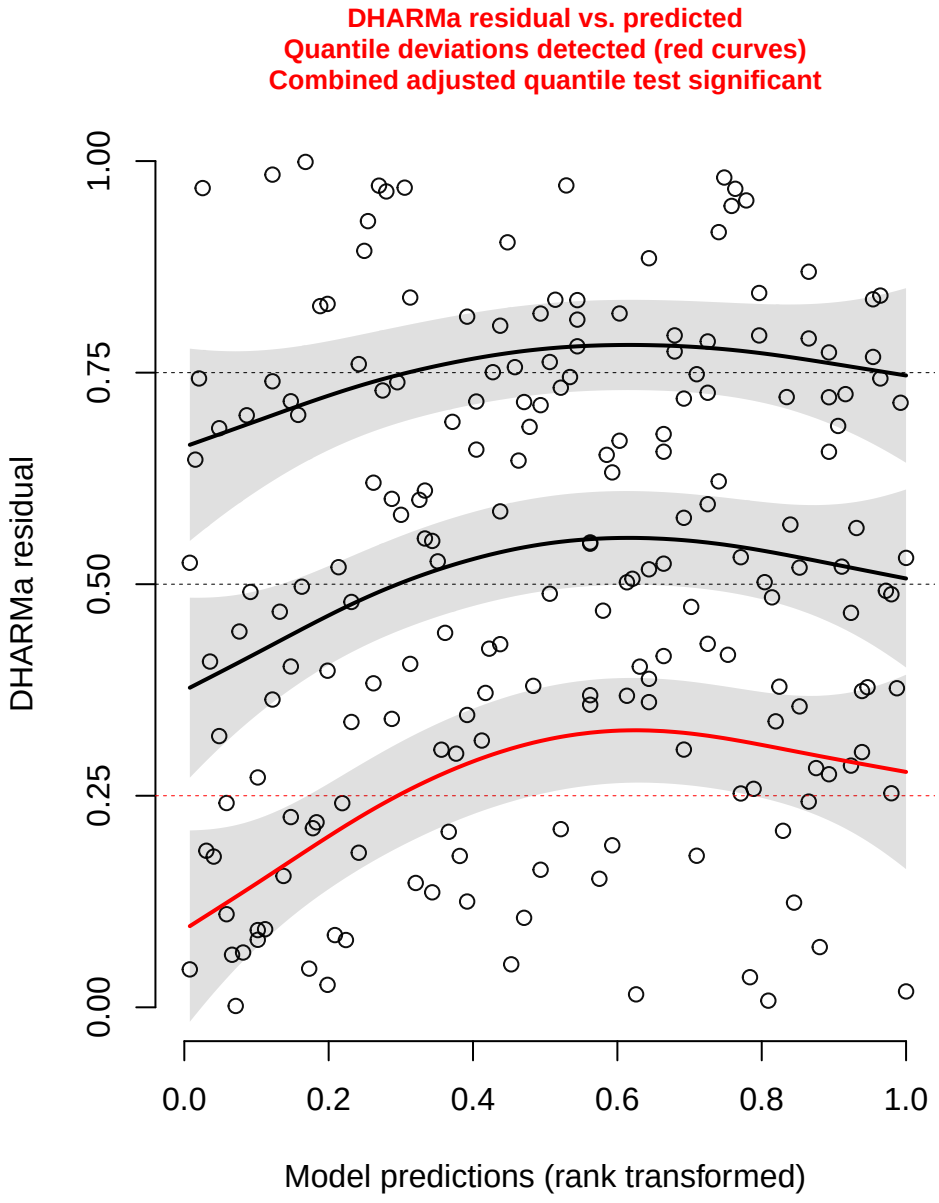
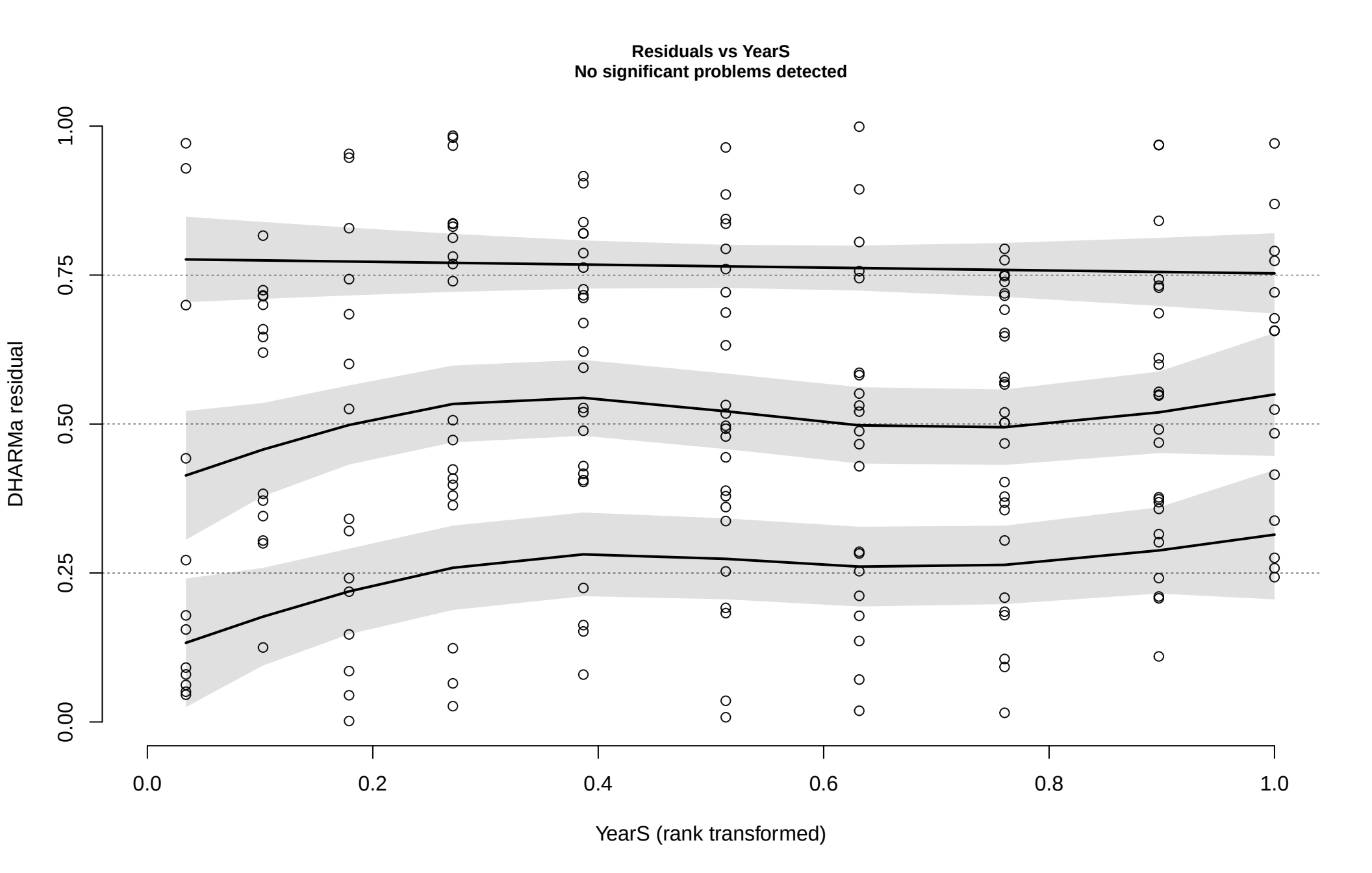
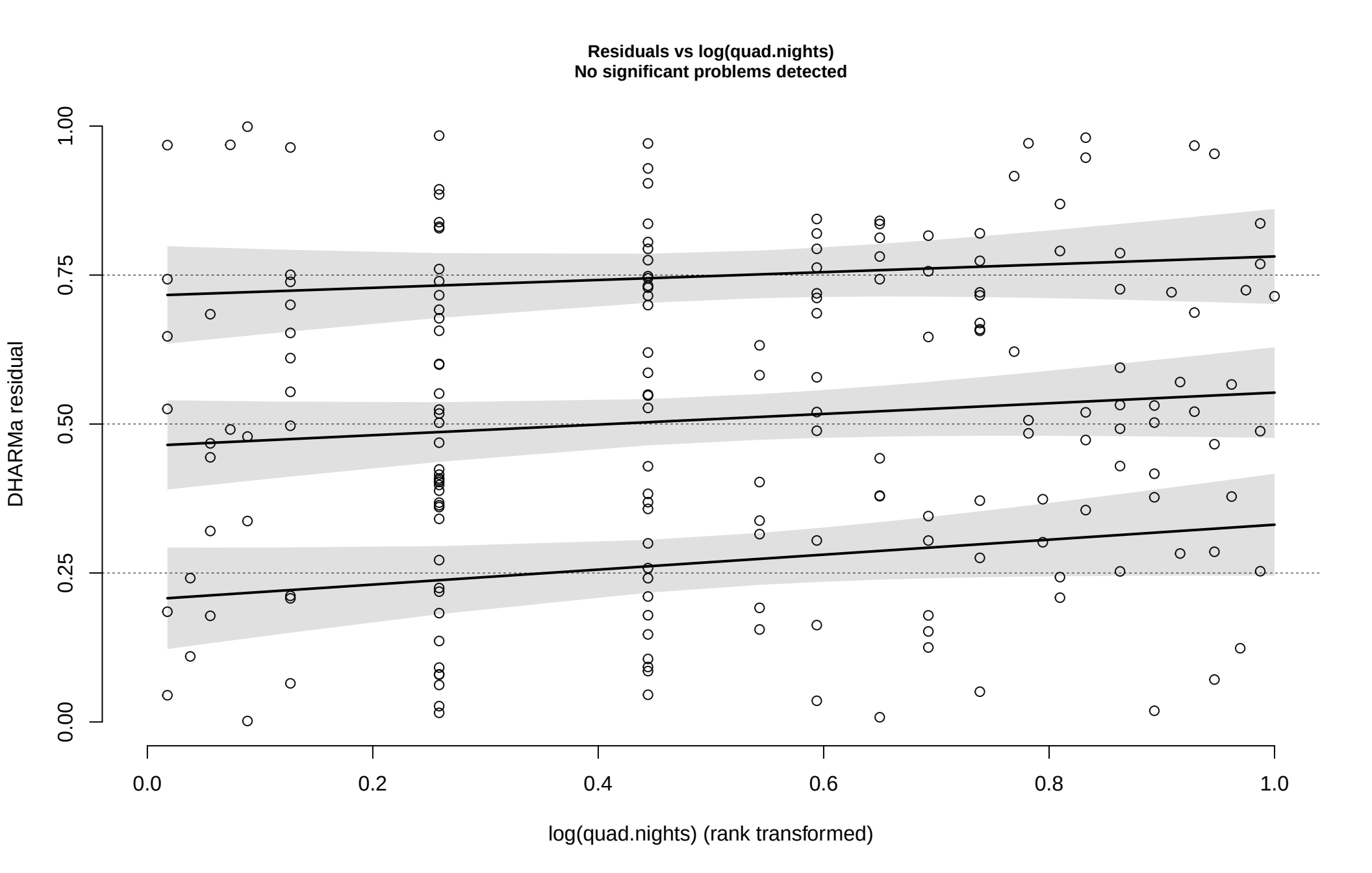
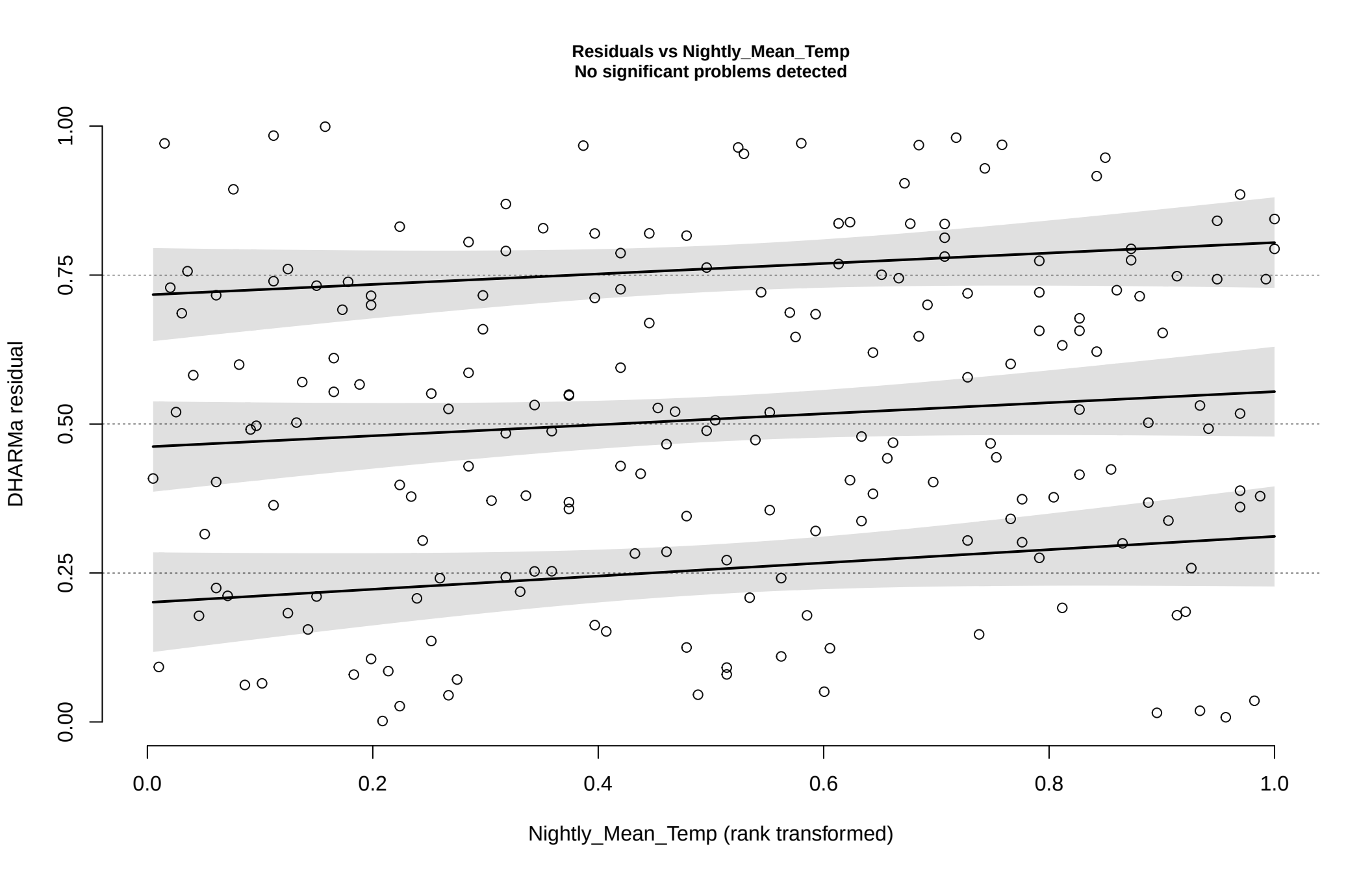
DHARMa residuals: ANPA

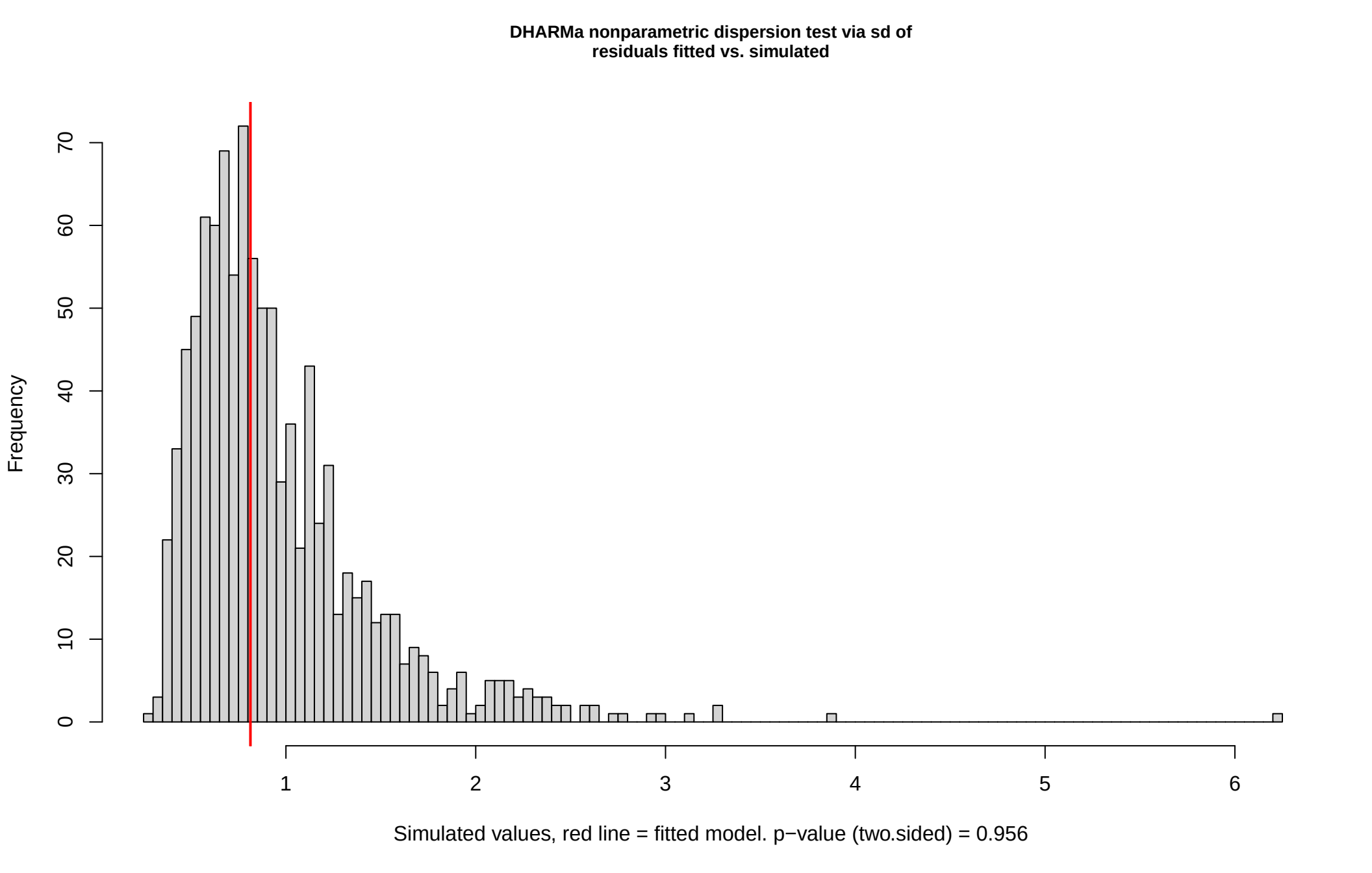


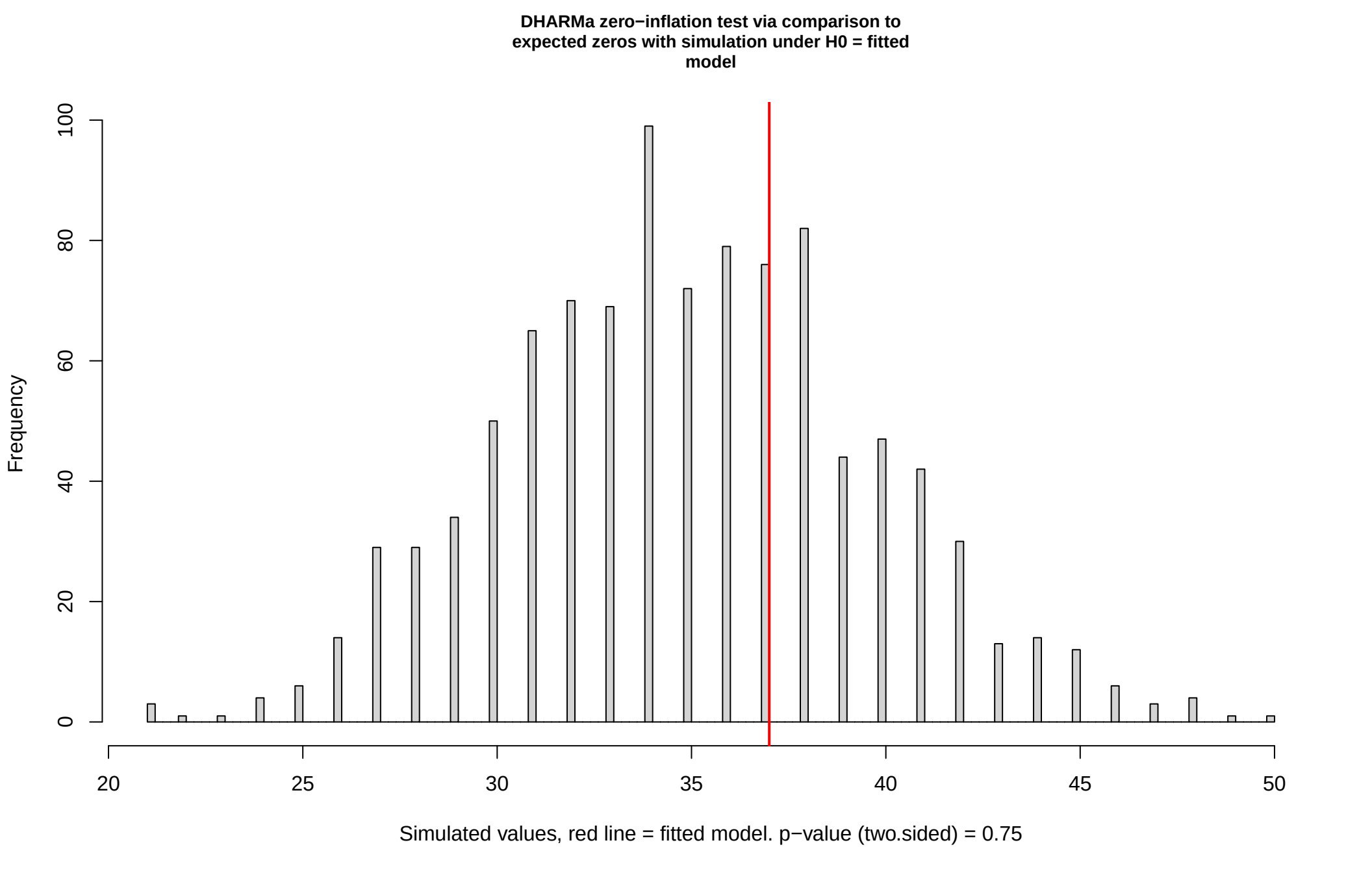


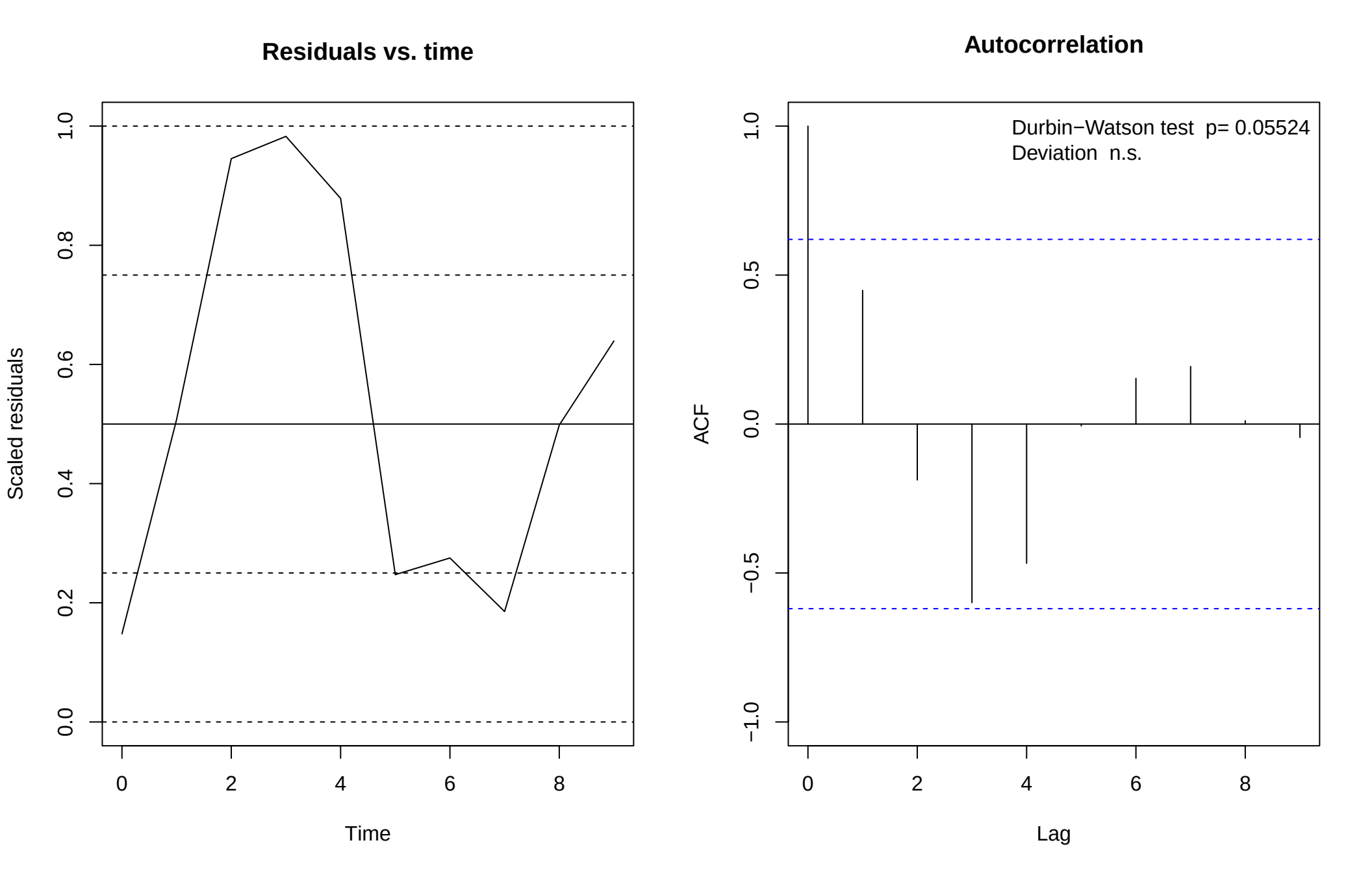

Residuals vs YearS
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

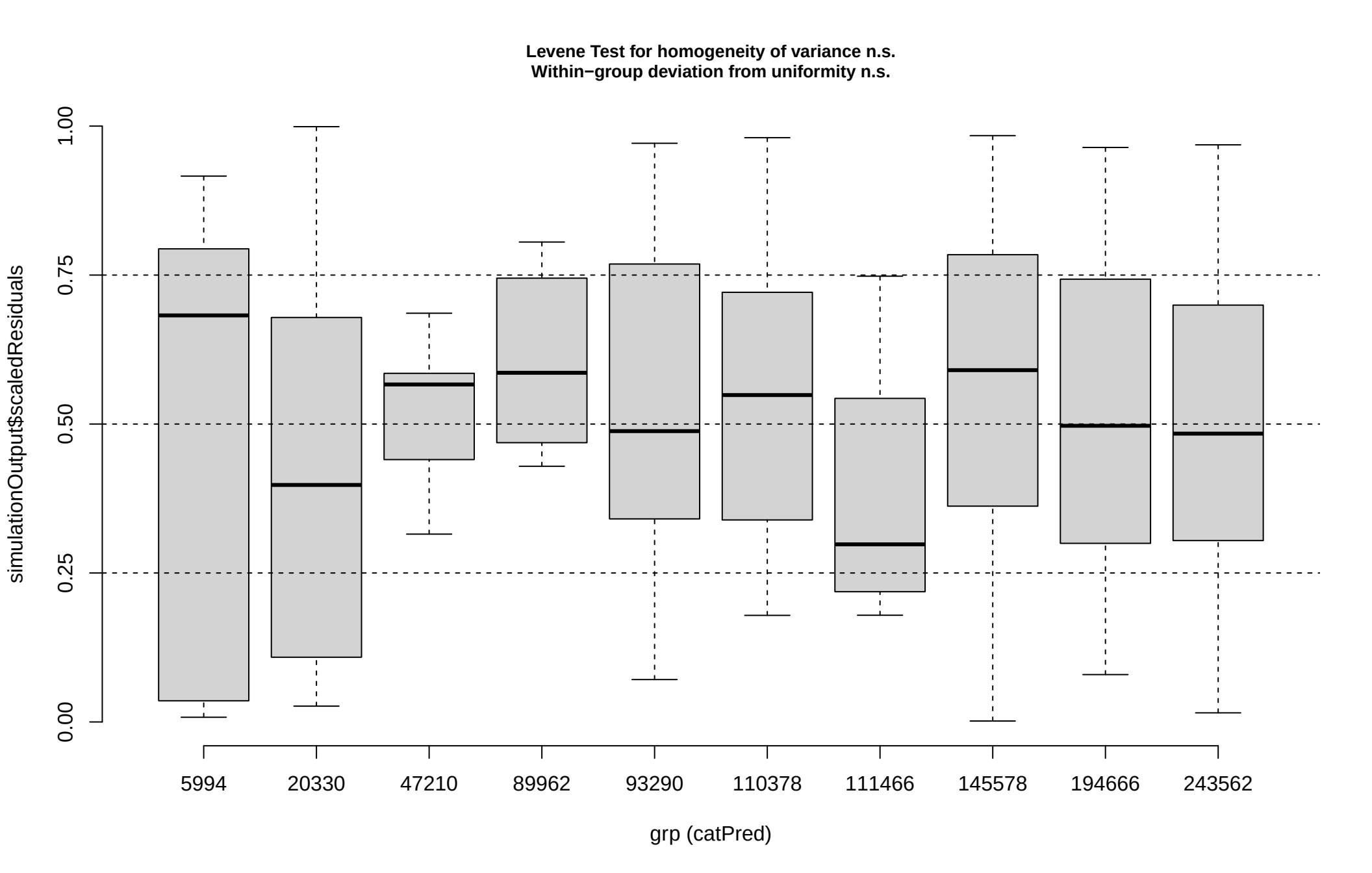

Residuals vs log(quad.nights)
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

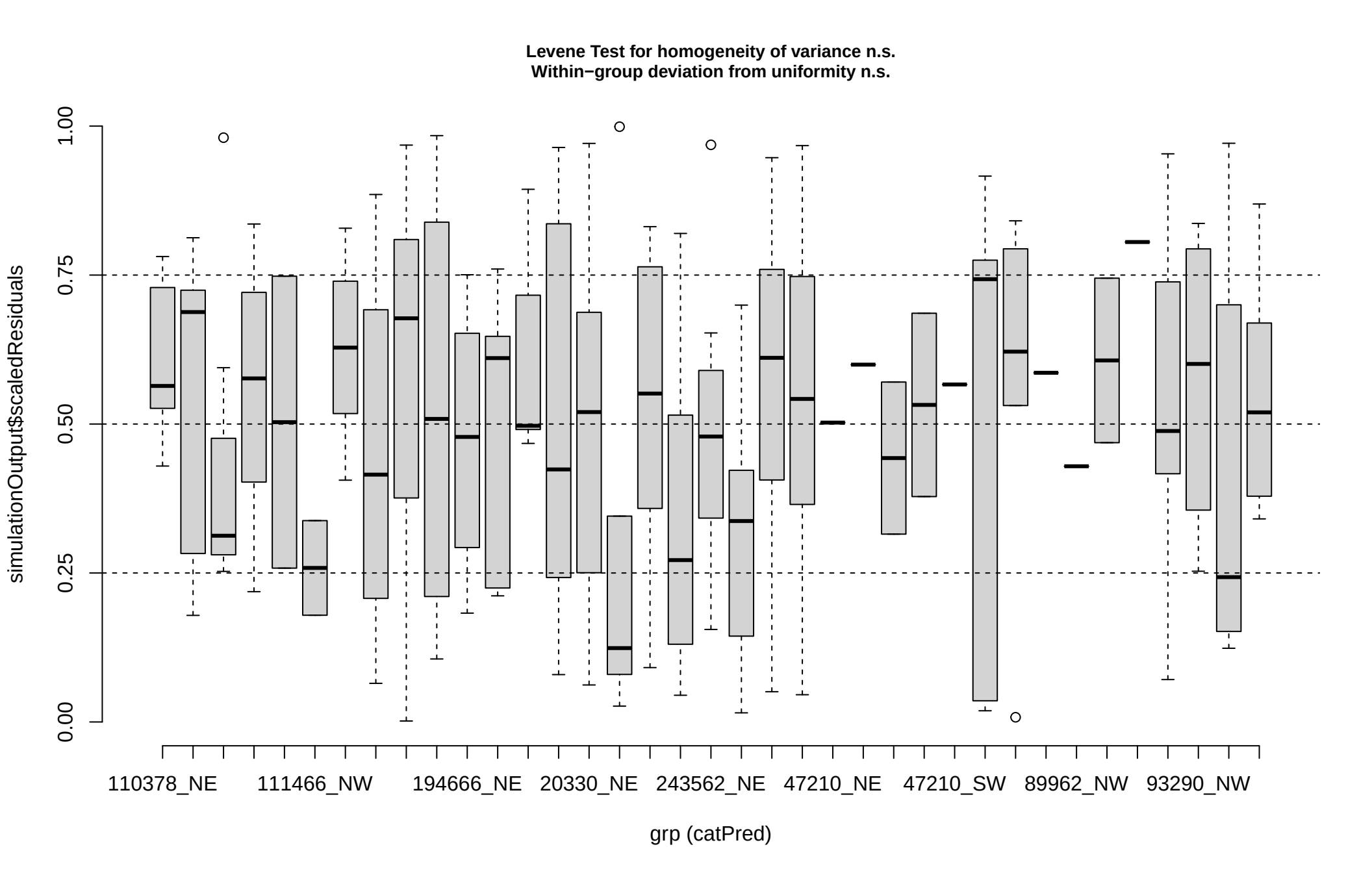

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

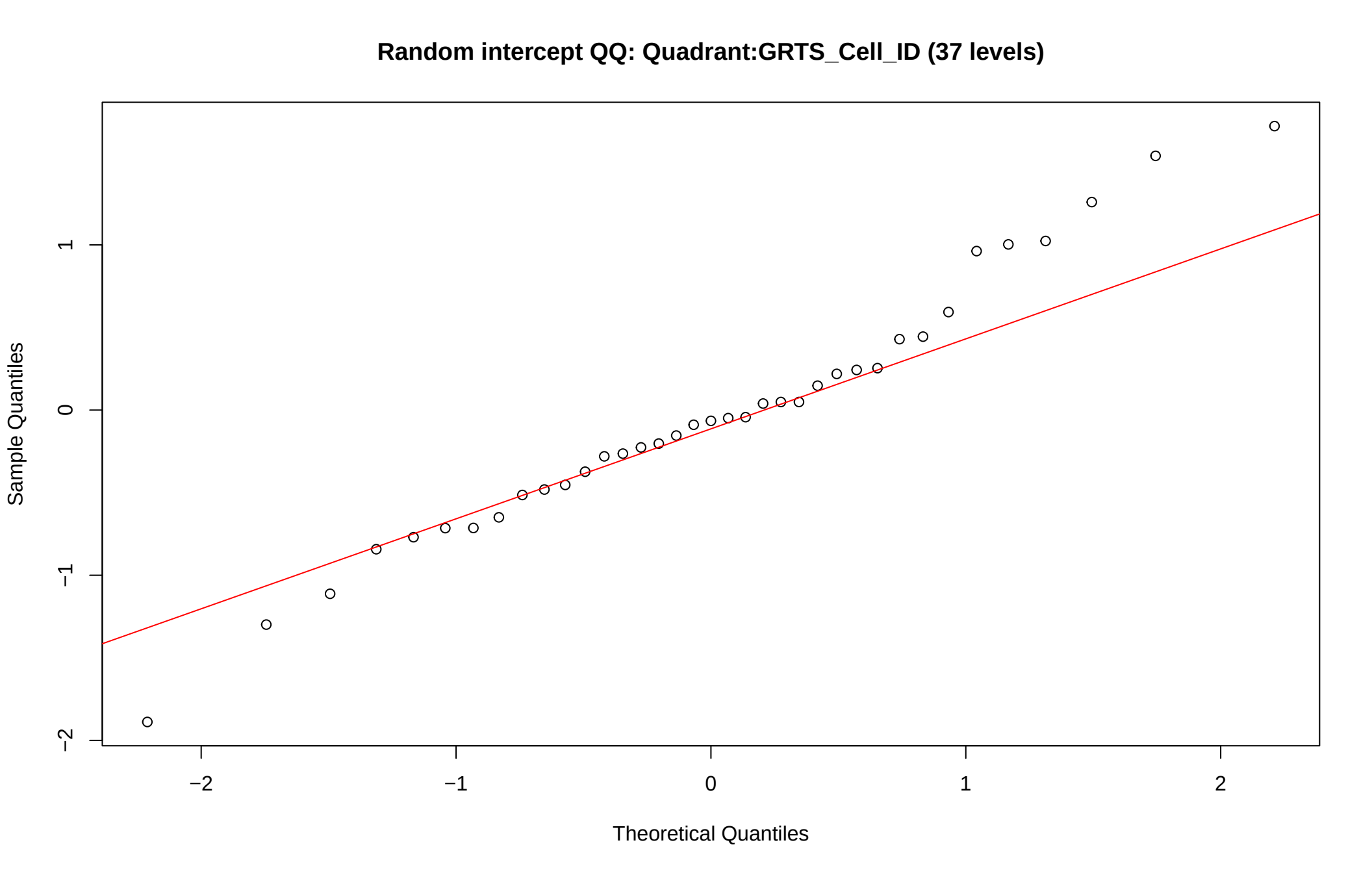

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
10
20
30
40
50
60
70
1
2
3
4
5
6
Simulated values, red line = fitted model. p−value (two.sided) = 0.956

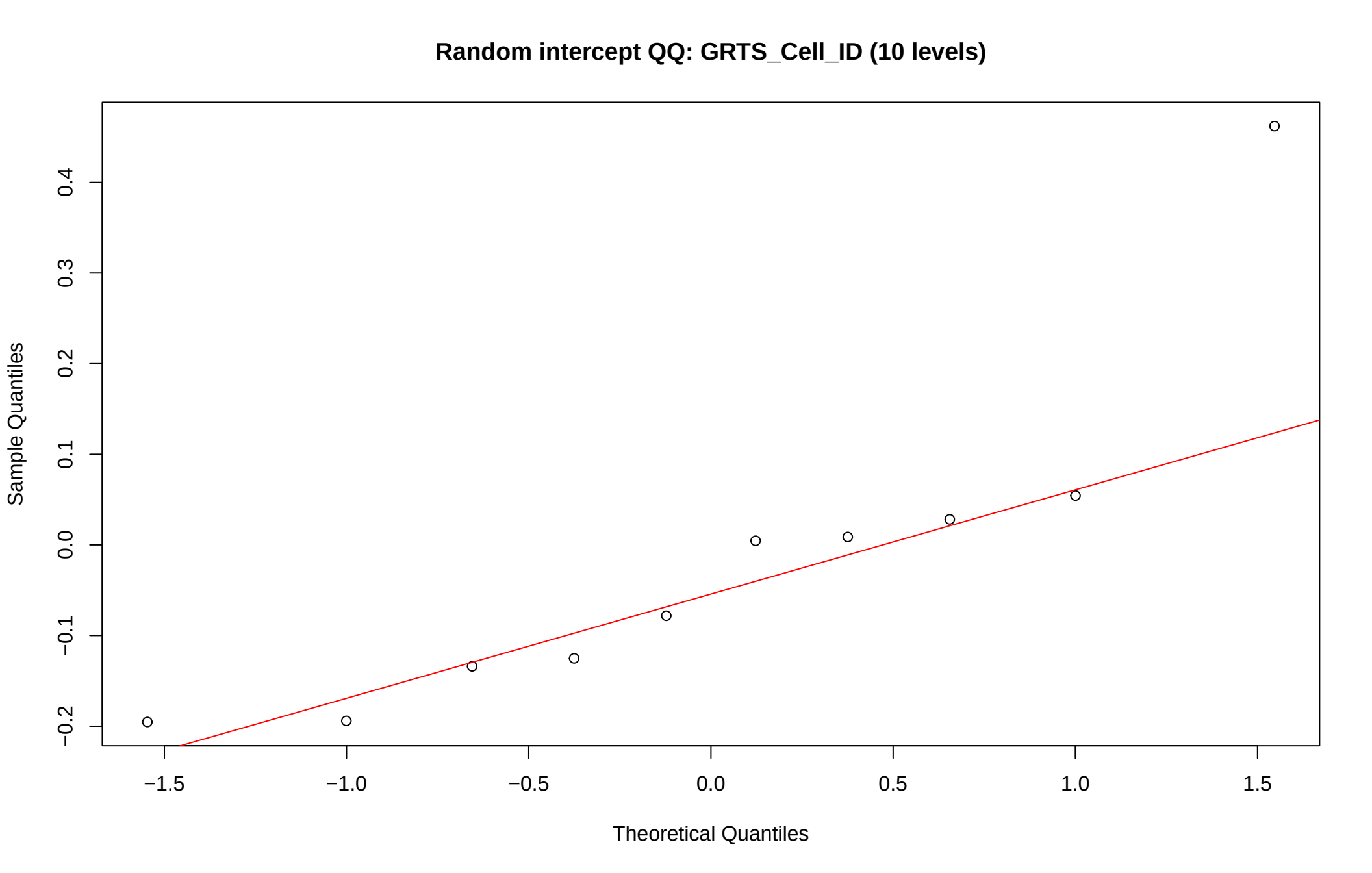

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
100
20
25
30
35
40
45
50
Simulated values, red line = fitted model. p−value (two.sided) = 0.75


Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
Time
Autocorrelation
ACF
−1.0
−0.5
0.0
0.5
1.0
Durbin−Watson test p= 0.05524
Deviation n.s.
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
5994
20330
47210
89962
93290
110378
111466
145578
194666
243562
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
110378_NE
111466_NW
194666_NE
20330_NE
243562_NE
47210_NE
47210_SW
89962_NW
93290_NW
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (37 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
−2
−1
0
1


Random intercept QQ: GRTS_Cell_ID (10 levels)
Sample Quantiles
Theoretical Quantiles
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5